



CSE-476: Introduction to Natural Language Processing
Quiz 3

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

- Q1. What stores the final output of the entire sequence?
- (A) To store the final output of the entire sequence.
 - (B) To act as a memory that carries information from previous time steps
 - (C) To calculate the loss function for the current word
 - (D) To normalize the input embeddings
- Q2. What prevents the gradients from exploding?
- (A) To ensure the output values sum to 1, representing a probability distribution
 - (B) To prevent the gradients from exploding
 - (C) To increase the speed of the training process
 - (D) To allow the model to handle variable-length sequences
- Q3. Which networks cannot use embeddings?
- (A) They cannot use embeddings
 - (B) They assume fixed input size and struggle with temporal order
 - (C) They cannot be trained with gradient descent
 - (D) They do not use activation functions
- Q4. What updates weights based on the loss?
- (A) To update weights based on the loss
 - (B) To calculate the predicted output and compute the loss
 - (C) To unroll the network into a sequence
 - (D) To calculate the gradient for each parameter
- Q5. When are weights re-initialized at every step?
- (A) They are randomly re-initialized at every step
 - (B) They are reused across timestamps
 - (C) They increase in value as the sequence progresses

- (D) Only the output weight V is reused
- Q6. What is equivalent to a deeper network with more representations?
- (A) A deeper network with more representational power
 - (B) A single linear model
 - (C) A recurrent neural network
 - (D) A decision tree
- Q7. When does the model predict the next word given predicted history?
- (A) The next word given the predicted history
 - (B) The next word given the ground-truth history
 - (C) The current word given itself
 - (D) All words simultaneously
- Q8. When does it depend only on the input at time t ?
- (A) Only the input at time t
 - (B) Only the hidden state at time $t - 1$
 - (C) Both the input at time t and the hidden state at time $t - 1$
 - (D) All future inputs
- Q9. For $h = f(Wx)$ with x of shape $(n_0 \times 1)$ and h of shape $(n_1 \times 1)$
- (A) $(n_0 \times n_1)$
 - (B) $(n_1 \times n_0)$
 - (C) $(n_0 \times 1)$
 - (D) $(1 \times n_1)$
- Q10. What computes predictions during inference?
- (A) To compute predictions during inference
 - (B) To calculate gradients of the loss with respect to parameters
 - (C) To initialize weights randomly
 - (D) To normalize input data

End of Paper